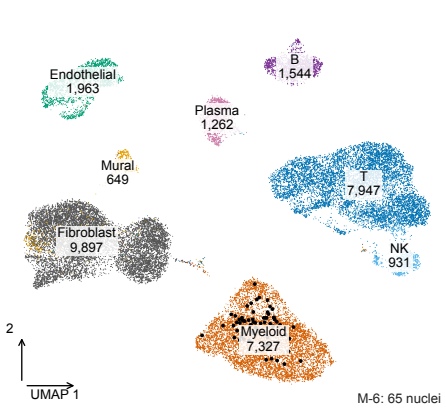
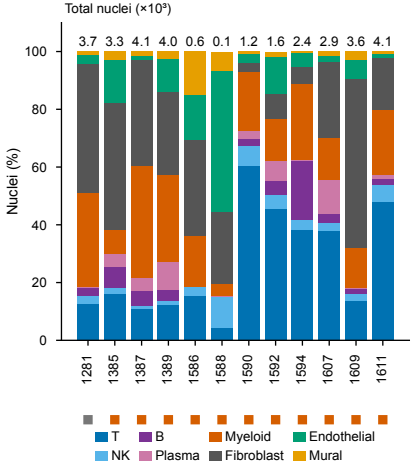
a **Study cohorts and evidence layers**

Genetics	Joint RNA + ATAC	Macrophages	Bulk synovium	Spatial tissue	Protein / context
3 RA GWAS	31,520 nuclei	26 donors	218 biopsies	17 Xenium sections	24 LCM regions
EUR · EAS · FinnGen	12 donors	Healthy · active · remission	46 paired anti-TNF patients	12 Visium sections	Blood · lung cancer
S-LDSC · PIP · QTL · MR	GSE243917	E-MTAB-8322	GSE89408 · GSE198520	Matched tissue blocks	Targets · perturbations

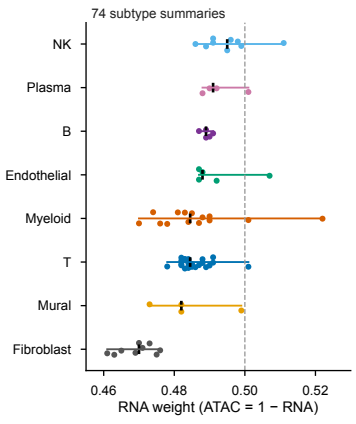
b **Joint multimodal atlas**



c **Donor composition and yield**



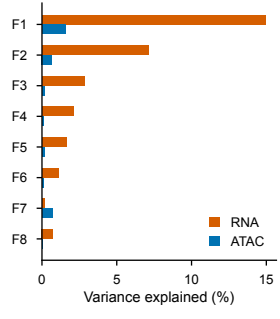
d **Modality contribution by state**



e **Integration benchmark: biology, mixing and rare-state recovery**

	Cell type ARI	Subtype ARI	ATAC ARI	Sample mixing	M-6 purity	M-6 F1	Overall
Concat. + Harmony	0.95	0.42	0.75	0.81	0.19	0.04	0.84
WNN	0.83	0.41	0.73	0.82	0.09	0.04	0.71
ATAC + Harmony	0.88	0.37	0.83	0.87	0.10	0.04	0.66
RNA + Harmony	0.81	0.41	0.64	0.83	0.24	0.04	0.54
RNA uncorrected	0.79	0.29	0.60	0.42	0.19	0.04	0.29
MOFA+ + Harmony	0.83	0.24	0.61	0.86	0.07	0.03	0.26

f **Joint latent-factor variance**



g **Donor RNA / ATAC quality (size: nuclei)**

